ALICE L WALTON SCHOOL OF MEDICINE

Gastrointestinal Mucosal Immunity & Immune Failures

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Learning Objectives (5)

After completing this brick, you will be able to:

1. Define the key structural components of gut-associated lymphoid tissue
2. Describe the innate and adaptive immune defenses maintaining GI mucosal homeostasis
3. Explain the role of intestinal microbiota and cytokine balance in regulating tolerance versus immune activation
4. Apply knowledge of GI mucosal immunity to clinical cases of celiac disease, IBD, and *C. difficile* infection
5. Analyze how failures in GI mucosal immune mechanisms contribute to specific disease manifestations

CASE CONNECTION

AS, a 24-year-old woman, presents to a GI outpatient clinic with several months of foul-smelling diarrhea, weight loss, and fatigue. Her symptoms worsen after eating pasta and bread. Physical exam reveals pallor and mild abdominal distension. Labs show iron-deficiency anemia, and a duodenal biopsy demonstrates villous atrophy with crypt hyperplasia and increased intraepithelial lymphocytes. JD, a second-year medical student on internal medicine rotation, is observing the evaluation of patients with chronic diarrhea and GI mucosal lesions.

What structural features of GI-associated lymphoid tissue are most relevant to these cases? How do immune mechanisms in the gut provide protection against such conditions? Consider your answers as you read, and we'll revisit AS at the end of the brick.

GO TO CONCLUSION ↓

What Are the Key Structural Components of Gut–Associated Lymphoid Tissue? (LO 1)

The gastrointestinal tract hosts the largest immune tissue network in the body, collectively called gut-associated lymphoid tissue (GALT). GALT must continuously distinguish between harmless dietary antigens and commensal bacteria on one hand, and genuine pathogens on the other. This remarkable feat depends on a coordinated architecture of specialized cells and structures distributed throughout the intestinal wall, each contributing a distinct function to mucosal defense and immune surveillance.

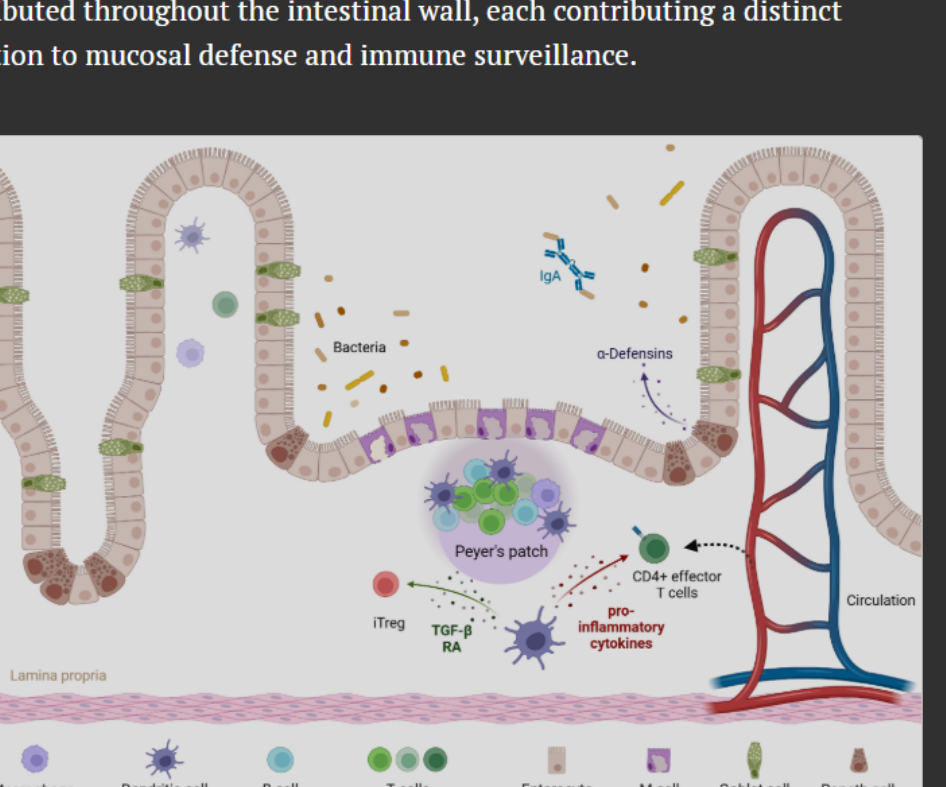

Figure 1: The Immune System of the Gut

Peyer Patches and [illegible]

As previously introduced, Peyer (or: Peyer's) patches (PPs) are organized lymphoid follicles located predominantly in the submucosa of the distal small intestine, particularly the ileum **(see Fig. 2)**. They serve as primary sites for antigen sampling and the initiation of adaptive immune responses within the gut. B cells residing in PPs undergo IgA class switching, a process driven by the cytokine TGF-β. Once switched, these B cells differentiate into plasma cells that migrate to the lamina propria, where they secrete dimeric IgA. This IgA is then transported across the epithelium into the intestinal lumen via a process called **transcytosis**, forming secretory IgA (sIgA) **(see Fig. 3)**. The follicular architecture of PPs—with germinal centers, T-cell zones, and overlying specialized epithelium—makes them the organizational hub of intestinal adaptive immunity.

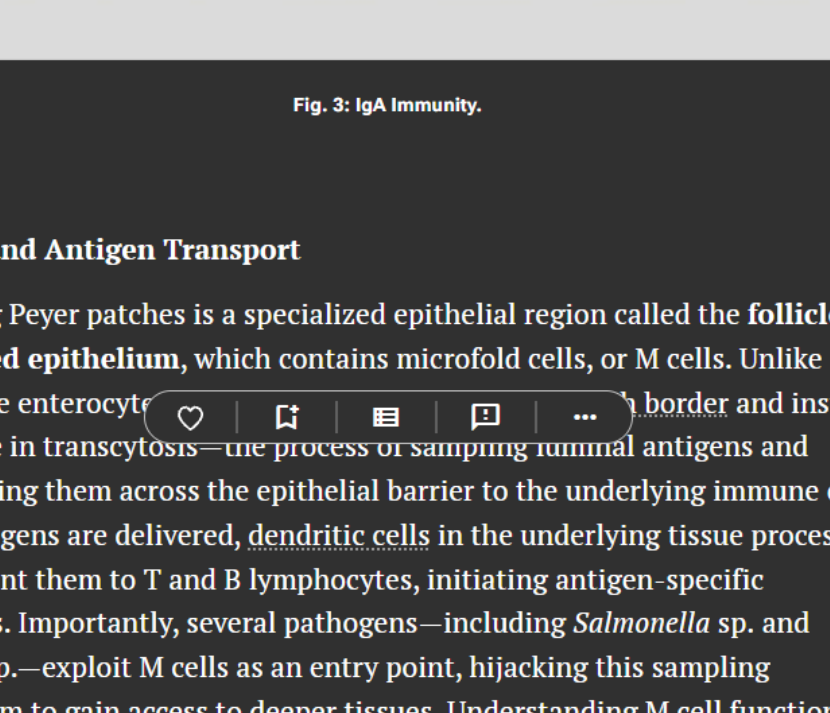

Fig. 2: Peyer Patches (PPs)

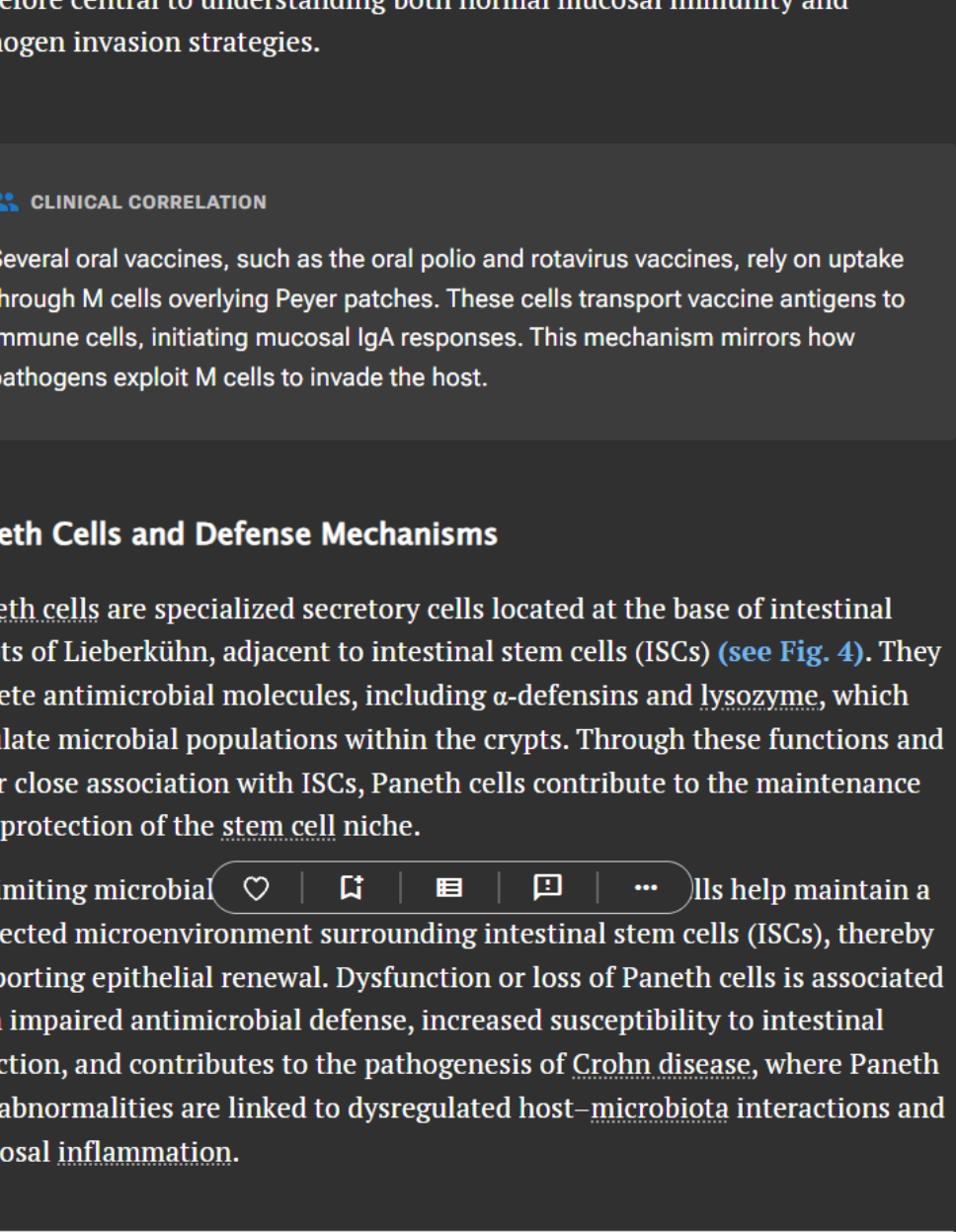

Fig. 3: IgA Immunity.

M Cells and Antigen Transport

Overlying Peyer patches is a specialized epithelial region called the **follicle-associated epithelium**, which contains microfold cells, or M cells. Unlike absorptive enterocyte[illegible] border and instead specialize in transcytosis—the process of sampling luminal antigens and transporting them across the epithelial barrier to the underlying immune cells. Once antigens are delivered, dendritic cells in the underlying tissue process and present them to T and B lymphocytes, initiating antigen-specific responses. Importantly, several pathogens—including *Salmonella* sp. and *Shigella* sp.—exploit M cells as an entry point, hijacking this sampling mechanism to gain access to deeper tissues. Understanding M cell function is therefore central to understanding both normal mucosal immunity and pathogen invasion strategies.

CLINICAL CORRELATION

Several oral vaccines, such as the oral polio and rotavirus vaccines, rely on uptake through M cells overlying Peyer patches. These cells transport vaccine antigens to immune cells, initiating mucosal IgA responses. This mechanism mirrors how pathogens exploit M cells to invade the host.

Paneth Cells and Defense Mechanisms

Paneth cells are specialized secretory cells located at the base of intestinal crypts of Lieberkühn, adjacent to intestinal stem cells (ISCs) **(see Fig. 4)**. They secrete antimicrobial molecules, including α-defensins and lysozyme, which regulate microbial populations within the crypts. Through these functions and their close association with ISCs, Paneth cells contribute to the maintenance and protection of the stem cell niche.

By limiting microbial [illegible]lls help maintain a protected microenvironment surrounding intestinal stem cells (ISCs), thereby supporting epithelial renewal. Dysfunction or loss of Paneth cells is associated with impaired antimicrobial defense, increased susceptibility to intestinal infection, and contributes to the pathogenesis of Crohn disease, where Paneth cell abnormalities are linked to dysregulated host–microbiota interactions and mucosal inflammation.

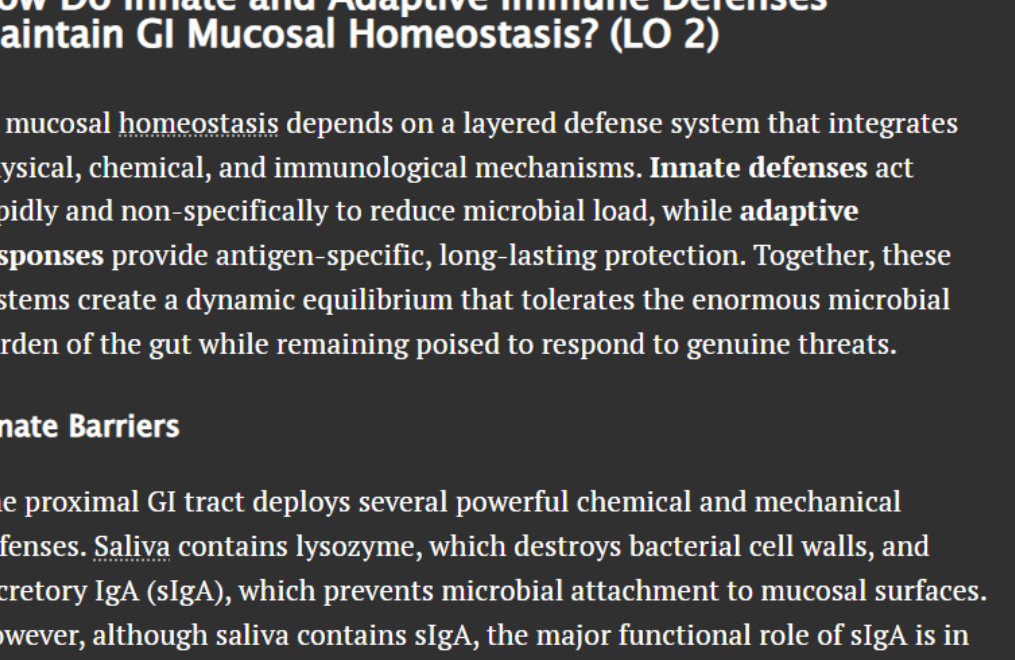

Fig. 4: Paneth cells

What are the core roles of Peyer patches, M cells, and Paneth cells in GI immunity?

How Do Innate and Adaptive Immune Defenses Maintain GI Mucosal Homeostasis? (LO 2)

GI mucosal homeostasis depends on a layered defense system that integrates physical, chemical, and immunological mechanisms. **Innate defenses** act rapidly and non-specifically to reduce microbial load, while **adaptive responses** provide antigen-specific, long-lasting protection. Together, these systems create a dynamic equilibrium that tolerates the enormous microbial burden of the gut while remaining poised to respond to genuine threats.

Innate Barriers

The proximal GI tract deploys several powerful chemical and mechanical defenses. Saliva contains lysozyme, which destroys bacterial cell walls, and secretory IgA (sIgA), which prevents microbial attachment to mucosal surfaces. However, although saliva contains sIgA, the major functional role of sIgA is in the intestinal lumen.

In the stomach, hydro[illegible] 1–2, killing the vast majority of ingested [illegible] intestine secrete mucus that traps microorganisms, while tight junctions between epithelial cells prevent paracellular leakage of luminal contents. Peristalsis continuously propels contents distally, physically clearing pathogens before they can establish colonization. The clinical importance of these barriers is illustrated by the increased infection risk seen with proton pump inhibitor (PPI) use or achlorhydria, which eliminates the gastric acid barrier, increasing the survival of ingested pathogens. While mucosal defenses such as sIgA remain intact, the higher infectious inoculum reaching the intestine exceeds these defenses, predisposing patients to infections with organisms such as *Salmonella* sp., *Vibrio* sp., and *Campylobacter* sp.

CLINICAL CORRELATION

Chronic use of proton pump inhibitors or achlorhydria increases the risk of enteric infections by eliminating the gastric acid barrier that normally kills ingested pathogens. As a result, a higher infectious inoculum reaches the intestine and can overwhelm intact mucosal defenses such as secretory IgA. This predisposes patients to infections with organisms such as *Salmonella*, *Vibrio*, and *Campylobacter*.

Adaptive Responses

The adaptive arm of GI immunity centers on sIgA and intraepithelial lymphocytes (IELs). Secretory IgA is a dimeric antibody that is produced by plasma cells in the lamina propria and transported into the intestinal lumen via transcytosis thro[illegible]eutralizes pathogens without triggering complement activation or significant inflammation, making it ideally suited to the tolerogenic environment of the gut. IELs, which are predominantly CD8+ T cells residing within the epithelial layer, provide rapid, localized cytotoxic responses against virally infected or stressed epithelial cells. Together, sIgA and IELs form a complementary adaptive defense: sIgA prevents pathogen attachment and invasion, while IELs eliminate compromised cells before infection can spread.

ACTIVE LEARNING QUESTION

A patient with achlorhydria develops *Salmonella* gastroenteritis. Which defense was most critically impaired, increasing susceptibility?

A. ○ Peristalsis and lysozyme
B. ○ Gastric acid
C. ○ Tight junctions and mucus
D. ○ Paneth cell defensins
E. ○ Intraepithelial lymphocytes

Explanation (requires correct answer)

How Do Intestinal Microbiota and Cytokine Balance Regulate Tolerance Versus Immune Activation? (LO 3)

The intestinal immune system faces a fundamental challenge: it must remain tolerant of trillions of commensal microorganisms and harmless dietary antigens while retaining the capacity to mount vigorous responses against pathogens. This balance is not passive—it is actively maintained through complex interactions between the microbiota, epithelial cells, dendritic cells, and regulatory T cells (Tregs), all coordinated by a carefully calibrated cytokine environment.

Microbiota and Colonization Resistance

As introduced previously, the healthy intestinal microbiota is dominated by two major phyla: *Bacteroidetes* and *Firmicutes*. These commensals occupy ecological niches along the intestinal mucosa, competing with potential pathogens for nutrients and attachment sites—a phenomenon known as **colonization resistance.** Beyond physical competition, commensal bacteria produce short-chain fatty acids (SCFAs), particularly butyrate, through fermentation of dietary fiber. Butyrate serves as the primary energy source for colonocytes and also exerts potent immunomodulatory effects, promoting the differentiation of Tregs and reinforcing the tolerogenic tone of the intestinal immune system. When antibiotic use disrupts this microbial community—a state called **dysbiosis**—colonization resistance is lost, creating an opportunity for opportunistic pathogens such as *Clostridioides difficile* to overgrow and cause disease.

CLINICAL CORRELATION

Diets low in fiber reduce the production of short-chain fatty acids such as butyrate by gut microbiota. This can impair Treg induction and shift the mucosal immune balance toward pro-inflammatory responses, contributing to diseases such as IBD.

Cytokine Profiles and Treg Induction

Distinct cytokine profiles orchestrate the balance between immune tolerance and activation in the gut. Under homeostatic conditions, tolerogenic dendritic cells produce TGF-β and retinoic acid, promoting differentiation of helper T cells into regulatory T cells (Tregs) and suppressing effector T-cell responses. This tolerogenic environment is essential for oral tolerance—the process by which repeated exposure to luminal antigens (food proteins, commensals) induces antigen-specific unresponsiveness rather than inflammation. During infection or immune dysregulation, this balance shifts toward a pro-inflammatory cytokine milieu. IL-12 and interferon-γ (IFN-γ) drive Th1 cell differentiation, while IL-1, IL-6, TGF-β, and IL-23 promote Th17 differentiation. Th1 and Th17 cells produce IFN-γ and IL-17, respectively, which enhance inflammation, phagocyte recruitment, and pathogen clearance.

The table below summarizes the key features of tolerance versus immune activation:

Table

Feature	[illegible]	[illegible] Activation
Trigger	Food antigens, commensals (homeostatic signals)	Pathogens, dysbiosis (danger signals)
Key cells	Tregs	Th1 / Th17 effector cells
Cytokines	IL-10, TGF-β	IFN-γ, IL-17
Outcome	Suppression, homeostasis	Inflammation, pathogen clearance

Which cytokines and microbial signals promote immune tolerance in the gut?

CLINICAL CORRELATION

Immune checkpoint inhibitors (e.g., anti–PD-1 therapy) used in cancer treatment can disrupt Treg-mediated tolerance and trigger colitis resembling inflammatory bowel disease. This [illegible]thways in preventing excessive [illegible]

How Can Knowledge of GI Mucosal Immunity Be Applied to Clinical Cases of Celiac Disease, IBD, and *C. Difficile* Infection? (LO 4)

Understanding the mechanisms of GI mucosal immunity becomes clinically powerful when applied to specific diseases. Celiac disease, inflammatory bowel disease (IBD), and *Clostridioides difficile* (*C. difficile*) infection each represent a distinct failure mode of the mucosal immune system—loss of oral tolerance, dysregulated effector responses, and microbiota disruption, respectively.

Celiac Disease

Celiac disease results from a breakdown of oral tolerance to gliadin, a component of gluten found in wheat, barley, and rye. In genetically susceptible individuals (typically HLA-DQ2/DQ8), gliadin peptides are presented by dendritic cells to CD4+ T cells, triggering a Th1-mediated inflammatory response in the small intestinal mucosa. This immune-mediated damage leads to villous atrophy and crypt hyperplasia, dramatically reducing the absorptive surface area of the small intestine. The resulting malabsorption affects iron, folate, calcium, and fat-soluble vitamins, producing the classic clinical picture of diarrhea, weight loss, and nutritional deficiencies. Increased IELs in the epithelium are a histological hallmark, reflecting the ongoing cytotoxic response to gliadin-expressing enterocytes.

IBD: Crohn and Ulcerative Colitis

Inflammatory bowel disease (IBD) encompasses two major conditions—Crohn disease (CD) and ulcerative colitis (UC)—both characterized by chronic, relapsing intestinal inflammation driven by dysregulated immune responses to commensal microbiota. In Crohn disease, a Th1-dominant response (mediated by IFN-γ and IL-12) produces transmural inflammation that can affect any segment of the GI tract, characteristically with skip lesions and non-caseating granulomas. Ulcerative colitis, by contrast, involves an atypical Th2 response that is limited to the colonic mucosa and submucosa, always involving the rectum and extending proximally in a continuous pattern. The distinction between transmural and mucosal inflammation has important clinical consequences: CD is associated with complications such as fistulas and strictures, while UC carries a higher risk of colorectal cancer with long-standing disease.

C. Difficile Infection

Clostridioides difficile infection (CDI) illustrates the critical importance of the intestinal microbiota as an immune defense. CDI typically follows antibiotic use, which disrupts the normal microbial community and eliminates colonization resistance. *C. difficile* spores, which are resistant to many antibiotics, germinate and proliferate in the disrupted environment. The organism produces Toxin A (enterotoxin) and Toxin B (cytotoxin), which damage the colonic epithelium, disrupt tight junctions, and trigger a robust inflammatory response. The result is pseudomembranous colitis, characterized by the formation of inflammatory exudates (pseudomembranes) on the colonic mucosa, severe diarrhea, and leukocytosis. Patients with impaired mucosal immunity—such as those on immunosuppressants or with hypogammaglobuline[illegible] severe or recurrent CDI.

ACTIVE LEARNING QUESTION

A patient with celiac disease continues eating gluten. Which mechanism best explains her persistent villous atrophy?

A. ○ IgE-mediated mast cell activation
B. ○ Neutrophil-mediated crypt damage
C. ○ CD4+ T-cell mucosal damage
D. ○ C. difficile toxin production
E. ○ Th2 eosinophilic inflammation

Explanation (requires correct answer)

CLINICAL CORRELATION

Remember: Fecal microbiota transplantation (FMT) restores microbial diversity in patients with recurrent *C. difficile* infection, re-establishing colonization resistance and suppressing pathogen overgrowth.

How Do Failures in GI Mucosal Immune Mechanisms Lead to Specific Disease Manifestations? (LO 5)

When the layered defenses of the GI mucosa fail—whether through structural breakdown, immune dysregulation, or microbial disruption—predictable clinical syndromes emerge. Analyzing these failures not only explains disease pathophysiology but also highlights potential therapeutic targets. Three major categories of immune failure underlie most GI immune-mediated diseases: loss of barrier function, disrupted immune regulation, and microbiota dysbiosis.

Loss of Barrier Function

The intestinal epithelial barrier depends on the integrity of tight junctions between enterocytes, a continuous mucus layer produced by goblet cells, and the antimicrobial peptides **(i.e. defensins** and **lysozymes)** secreted by Paneth cells. When any of these components fails, the result is increased intestinal permeability—colloquially termed "leaky gut." In celiac disease, immune-mediated destruction of enterocytes disrupts tight junctions, allowing luminal antigens to penetrate the lamina propria and amplify the inflammatory response. In IBD, chronic inflammation similarly degrades barrier integrity, creating a self-perpetuating cycle of antigen exposure and immune activation. Loss of mucus production or Paneth cell dysfunction further compounds susceptibility to microbial invasion and epithelial damage.

Disrupted Immune Regulation

Healthy GI immunity [illegible]reg-mediated suppression and effector [illegible] is disrupted—whether by genetic predisposition, environmental triggers, or microbial changes—the immune system may attack self-antigens or harmless luminal contents. In celiac disease, the failure of oral tolerance to gliadin reflects a breakdown in Treg-mediated suppression, allowing pathogenic CD4+ T-cell responses to proceed unchecked. In IBD, loss of IL-10 signaling or Treg function permits chronic Th1 or Th17 activation against commensal bacteria. The clinical consequences range from localized mucosal inflammation (UC) to transmural granulomatous disease (CD), depending on the specific effector pathways involved.

Microbiota Dysbiosis and Pathogen Overgrowth

Antibiotic-induced dysbiosis is a classic example of how microbiota disruption leads to disease. Broad-spectrum antibiotics eliminate not only pathogens but also the commensal bacteria that provide colonization resistance. The resulting ecological vacuum allows *C. difficile*—whose spores persist in the environment and are resistant to many antibiotics—to germinate and overgrow. Beyond CDI, dysbiosis has been implicated in the pathogenesis of IBD, where reduced microbial diversity and loss of SCFA-producing bacteria may impair Treg induction and promote chronic inflammation. Patients with primary immunodeficiencies affecting IgA production or T-cell function are similarly vulnerable to recurrent enteric infections, underscoring the interconnection between immune competence and microbiota stability.

ACTIVE LEARNING QUESTION

A hospitalized [illegible]er 10 days of broad-spectrum antibiotics. Stool toxin assay is positive. Which immune failure is the primary driver?

A. ○ Loss of colonization resistance
B. ○ Deficient sIgA production
C. ○ Villous atrophy and malabsorption
D. ○ Th1-driven transmural inflammation
E. ○ Paneth cell defensin deficiency

Explanation (requires correct answer)

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to AS, what structural component of GALT is involved in sampling dietary antigens? How do defects in tolerance or barrier function contribute to her villous atrophy and malabsorption?

M cells overlying Peyer patches sample luminal antigens—including gliadin peptides—and deliver them to dendritic cells for presentation. In AS, a failure of oral tolerance allows CD4+ T cells to mount a Th1 response against gliadin, causing immune-mediated destruction of intestinal villi. This villous atrophy reduces absorptive s[illegible], iron-deficiency anemia, and weight [illegible]plify antigen exposure, perpetuating the inflammatory cycle. A strict gluten-free diet removes the antigenic trigger, allowing mucosal healing.

Summary

- GALT is the largest immune tissue network in the body, organized around Peyer patches (IgA class switching), M cells (antigen transcytosis), and Paneth cells (defensin/lysozyme secretion protecting intestinal stem cells).
- Innate GI defenses include gastric HCl, mucus, lysozyme, peristalsis, and tight junctions; loss of any component—such as acid suppression with PPIs—significantly increases infection susceptibility.
- Adaptive immunity in the gut relies on sIgA (non-inflammatory pathogen neutralization) and CD8+ intraepithelial lymphocytes (rapid cytotoxic responses), working together to provide antigen-specific mucosal protection.
- The intestinal microbiota (dominated by Bacteroidetes and Firmicutes) provides colonization resistance and produces SCFAs like butyrate, which induce Tregs and maintain immune tolerance; dysbiosis disrupts this balance.
- Immune tolerance is maintained by IL-10 and TGF-β promoting Treg activity; a shift to IFN-γ and IL-17 drives Th1/Th17 effector responses during infection or dysregulation.
- Celiac disease results from loss of oral tolerance to gliadin, causing CD4+ T-cell–mediated villous atrophy and malabsorption; IBD reflects chronic dysregulated effector responses (Th1 in Crohn, atypical Th2 in UC) against commensal bacte[illegible]
- C. difficile infection exemplifies microbiota dysbiosis: antibiotic use eliminates colonization resistance, enabling C. difficile overgrowth, toxin-mediated epithelial damage, and pseudomembranous colitis.

Review Questions

1. Which structure in GALT primarily initiates adaptive immune responses in the gut?

A. ○ Peyer patches
B. ○ Microfold (M) cells
C. ○ Paneth cells
D. ○ Goblet cells

Explanation (requires correct answer)

2. Which innate defenses contribute to mucosal homeostasis in the gastrointestinal tract?

A. ○ Lysozyme and sIgA
B. ○ Gastric acid and mucus
C. ○ Tight jun[illegible]
D. ○ Secretory IgA and IELs
E. ○ Paneth cell defensins

Explanation (requires correct answer)

3. Which cytokine is most important for the induction of regulatory T cells (Tregs) in the gut?

A. ○ IFN-γ
B. ○ IL-4
C. ○ TGF-β
D. ○ IL-17
E. ○ IL-12

Explanation (requires correct answer)

4. What immune mechanism underlies persistent villous atrophy in untreated celiac disease?

A. ○ IgE-medi[illegible]
B. ○ Neutrophil-mediated crypt damage
C. ○ CD4+ T-cell mucosal damage
D. ○ C. difficile toxin production
E. ○ Th2 eosinophilic inflammation

Explanation (requires correct answer)

5. Which failure chiefly allows *C. difficile* overgrowth after antibiotics?

A. ○ Loss of colonization resistance
B. ○ Deficient sIgA production
C. ○ Villous atrophy and malabsorption
D. ○ Th1-driven transmural inflammation
E. ○ Paneth cell defensin deficiency

Explanation (requires correct answer)

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☐ Define the key structural components of gut-associated lymphoid tissue
02 ☐ Describe the innate and adaptive immune defenses maintaining GI mucosal homeostasis
03 ☐ Explain the role of intestinal microbiota and cytokine balance in regulating tolerance versus immune activation
04 ☐ Apply knowledge of GI mucosal immunity to clinical cases of celiac disease, IBD, and *C. difficile* infection
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Objective Confidence: 0/5